

Ransomware Incident Brief

Overview

Summarized several incidents using the 5 W's framework to clearly communicate cause, impact, and scope to non-technical stakeholders.

Date: March 27, 2026	Entry: #1 - Ransomware
Description	Ransomware attack disrupted a healthcare clinic, initiated through a phishing email with a malicious attachment.
Tool(s) used	None
The 5 W's	● Who: An organized group of unethical hackers ● What: Ransomware encrypting critical systems ● When: Tuesday 9:00 a.m., during business operations ● Where: Internal systems of a healthcare clinic, including patient records and other critical data ● Why: The attack was likely financially motivated, as a ransom was demanded for file recovery. Attackers gained access through a phishing email with a malicious attachment. Once opened, malware was installed, allowing ransomware to encrypt critical files.
Additional notes	This incident follows a common attack pattern in which phishing is used to deliver malware that enables ransomware deployment, resulting in significant operational disruption. The primary impact was loss of system availability, with potential risk to sensitive patient data. The absence of backups significantly increased the severity of the incident, as the organization had limited ability to restore systems without paying the ransom. Key lessons learned include the need for phishing awareness training, endpoint protection, and reliable backup and recovery processes to reduce both the likelihood and impact of similar incidents.

Network Traffic Analysis

Date: March 29, 2026	Entry: #3 - Network Anomaly
Description	Unusual network traffic patterns indicated potential unauthorized communication with external systems.
Tool(s) used	Wireshark
The 5 W's	● Who: Unknown external system communicating with an internal host ● What: Suspicious outbound network traffic inconsistent with normal operations, suggesting potential unauthorized data exchange ● When: Friday 2:15pm, Identified during routine traffic monitoring ● Where: Internal system communicating with external IP addresses ● Why: Abnormal traffic patterns suggested a compromised system initiating unauthorized communication, potentially for command-and-control or data transfer. The activity was likely part of a larger attack sequence.
Additional notes	Analysis of packet data revealed unexpected connections and protocol usage inconsistent with normal operations. This type of activity may indicate command-and-control communication or data exfiltration attempts. The primary risk was compromise of system integrity and confidentiality of data. Limited visibility into network traffic due to lack of centralized monitoring and logging delayed detection. Key lessons learned include the need for automated network monitoring with anomaly detection, and improved visibility into system communications to identify and respond to suspicious behavior.

Data Integrity Breach

Date: March 30, 2026	Entry: #2 - Compromised Credentials
Description	Unauthorized changes to customer records indicated a data integrity breach following compromised user credentials.
Tool(s) used	Database logs, SIEM (Splunk)
The 5 W's	● Who: Unknown (internal or external) attacker(s) using compromised employee credentials ● What: Unauthorized modification of customer data ● When: Identified Monday at 3:45pm, after discrepancies were reported in customer records ● Where: Internal database containing customer information ● Why: Compromised credentials were used to access the system and alter data, likely to disrupt operations or manipulate records
Additional notes	This incident highlights how credential compromise can impact data integrity, not just access. The primary impact was loss of trust in data accuracy, which can disrupt business operations and decision-making. Lack of monitoring for unusual data changes delayed detection. Key lessons learned include the need for access controls, logging and monitoring of data changes, and validation processes to detect unauthorized modifications.